

Supplementary Reproducibility Record

Risk-Controlled Cross-Domain Battery State-of-Health Estimation via Prefix-Causal Harm-Budget Projection

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1 Frozen artifact chain

Table 1 records selected SHA-256 identifiers that bind the prefix-causal implementation, nested development evaluation, unprotected comparator, prospectively locked BaSyTec external validation, and NASA workflow to the reported results. Original datasets remain governed by their respective repositories and licences.

Table 1: Selected SHA-256 identifiers for the causal method and two protocol-frozen external workflows.

Artifact	SHA-256
Prefix-causal projection implementation	ce7288a129c17114e1ca57432c6417beba7938d58db2b1fd0a87171c479eb54c
Nested development predictions	d7a9a8c06923841260d98526dff665562ca27624357a4d514f6bc75563c69534
Candidate-control implementation	0f6523700082cff38aa26cd05385d3cf3252d436ccdbc22821355e7029861873
Source-tuned unprotected comparator report	ed11518ea3e1eaa42148bf4421f0570d775074ae697260a8edf9fbac0798c252
BaSyTec final pre-freeze protocol	2e7c34cc3e56cbd5901d18b3989d18754d08e66e7448205197e5b6aa9658643a
BaSyTec header-only inventory	35c4fa46ad72f00f09a17282e0ed4a2c04ec03c33c852ea621fafc746ab6a656
Frozen BaSyTec predictions	2e27780968f909b0d1d1a05b2b90370e2735e99f3a8e8926b781150751b6f8f6
BaSyTec one-shot result report	eb6bf718810577fe6e91228301e7352236a991d9899b3002fc2ee2d5753ec2b0
BaSyTec independent statistical audit	7726061409e3a897fa2677c519a7a1af8e3ca1771fac384d8f0feed2c0682fc7
NASA pre-freeze protocol	37507d16ed8f13bd4aa378f3da3480dad691addf383c31b00ad613bb7397adea
Official NASA archive	82302a7db4fc1b34e0b6676326610438d43b816bdf11a69d1d012a464ef2f92e
Frozen NASA predictions	4d8264d94e224be12a5574cc403047e10276c78652d80a4841086f4e409ceaa9
Independent NASA scoring program	9894c1cbecae80e8a0712b812257ab474974ee3a3ea8df418e6306f94bdf40ba
NASA one-shot result report	d92be40ab5c34b6324667af549cfde34b15857d02e8c0c654579111854de48b1

2 Guarantee and workflow verification

Thirty historical V321 unit tests cover causal-state monotonicity, unit assimilation, prefix invariance, nearest-current-feasible projection, range and tube membership, the exact harm identity and harm-region equivalence, zero-budget impossibility, recursive feasibility and exact viability-kernel membership, asymmetric-loss contracts, time-varying feasibility and minimum viable budgets, invalid contracts, non-finite and shape-mismatched inputs, and cellwise order restoration. Four separate V377 unit tests cover the exact local proximal representation, protected-state and complete-operator prefix nonexpansiveness, and tightness of the unit Lipschitz constant, giving 34 focused PCHP tests without modifying the historical V321 file. Four synthetic NASA workflow tests, five BaSyTec header-family tests, and one synthetic MATLAB integration test verify field isolation and two-stage execution. The development, BaSyTec, and NASA reports additionally check maximum displacement, observed absolute-loss increase, trajectory monotonicity, independent-unit counts, frozen file identities, and aggregate reconstruction.

3 Retrospective budget-path audit

The budget-path analysis was protocol-locked before its computation but after all development-domain outcomes had been opened. Its single primary empirical estimand is the domain-equal mean of the budget-normalized integrated PCHP-minus-exact-budget-feasible-shift MAE difference. Per-budget intervals are descriptive. The original V361 validator recorded **NARROW** because one historical anchor compared the high-precision V327 value with a report JSON rounded to ten decimal places under a 10^{-12} tolerance. V361 was not modified or rerun. A separately frozen V362 audit verified the authoritative aggregate CSV, all twelve domain rows, the V326 source-only selection roster, and all non-anchor scientific gates. It classified the anchor failure as a serialization-precision false negative while retaining the retrospective label and the original V361 decision record.

Table 2: SHA-256 identifiers for the budget-path analysis and independent anchor audit.

Artifact	SHA-256
V361 analysis pre-freeze	7bc32a42d2c20e61471357126f6281e3a11ac976979d56056b68b243f6a6c4d6
V361 analysis program	bdd0a1aeaf715d25fe574feb52b1679551a608999ab49848a3ff8c6dfd701f79
V361 result report	6e8d09554f73f1df7cdf6cf8affdaaf1c13eec80b67778b5e58616038ec02781b
V361 domain AUC table	f3c1275be42e9de2d38a4d516c3518a3777bad5a0341f4d07a2852d28693c416
V362 audit pre-freeze	49b986a77346f28c80c3d024bae6caff58de469124c422db726945c72400ea56
V362 audit program	ca3e2be80c1e995daaaa90741ab6d6e07af94424a294d6b80e09f050b32640b5
V362 audit report	8e02eca68f7016d0ec08100195689e0129cccc90fc8add7b6ecc4a275b323608

4 Finite-sample domain sensitivity audit

Because the development estimands have only 12 independent complete-domain effects, V378 re-computed three paired comparisons using complementary small-sample procedures. For each comparison it enumerated all $2^{12} = 4,096$ sign assignments for the domain-equal mean, computed a Student- t interval, and reported exact paired Wilcoxon and direction-only sign tests. This audit was performed after the development outcomes had been opened. It is a sensitivity analysis, not a new confirmatory test, and it leaves the frozen estimands and percentile-bootstrap reports unchanged.

Table 3: Post-hoc finite-sample sensitivity over complete domains. Negative means favor PCHP. All test values are two-sided and descriptive.

Comparison	Mean	Student- t 95% interval	Sign-flip value	Sign-test value	W/T/L
PCHP vs protected baseline	-0.004862	[-0.007033, -0.002692]	0.000977	0.006348	11/0/1
PCHP vs candidate-free shift	-0.004773	[-0.008581, -0.000965]	0.016602	0.145996	9/0/3
Integrated budget-path AUC	-0.006942	[-0.011718, -0.002166]	0.007812	0.006348	11/0/1

The magnitude-sensitive calculations preserve all three negative domain-equal mean directions. The direction-only sign test for the candidate-free comparison does not fall below 0.05, so that result supports an average utility contribution with heterogeneous domain effects, not universal dominance or a claim that improvement is more likely than not in a randomly drawn new domain.

Table 4: SHA-256 identifiers for the V378 finite-sample audit.

Artifact	SHA-256
V378 audit program	50426e90264f15e1632d0fa8403be7b16c7ef09c9076a34be05897f1a0b0b578
V378 audit report	411007452ed8c095258f3949af959e2518dc1c63e648583e5ffe9dde3b265ae9

5 Untuned secondary-backbone portability audit

This protocol-locked retrospective audit replaced both primary ExtraTrees regressors with Ridge, HistGradientBoosting, or LightGBM while preserving the data, feature contract, weighting, outer-domain exclusions, harm budget, PCHP operator, and exact V326 outer-domain α schedule. No secondary-backbone hyperparameter or α tuning was permitted. Each backbone produced 601,932 post-reference predictions over the same 586 cells and 12 complete domains. The independent verifier recomputed the domain effects and all deterministic certificates from the saved record-level predictions and passed 196 named checks.

Table 5: Zero-additional-tuning portability results. Differences are PCHP minus the corresponding protected-state MAE; negative values favor PCHP. Intervals are Bonferroni-adjusted two-sided 98.33% percentile intervals from complete-domain bootstrap resampling.

Backbone	Protected MAE	PCHP MAE	Difference	Adjusted interval	W/T/L
Ridge	0.11192	0.10773	-0.00419	[-0.00677, -0.00131]	10/0/2
HistGradientBoosting	0.07674	0.07249	-0.00425	[-0.00633, -0.00203]	11/0/1
LightGBM	0.07674	0.07252	-0.00422	[-0.00621, -0.00231]	11/0/1

All three pre-frozen conjunction gates passed. Maximum displacement and observed absolute-loss increase remained within the 0.01 budget, maximum protected-state and output increases were zero, outputs remained in $[0, 1.3]$, and exact prefix replay had maximum difference zero. Because all development outcomes had already been opened, this result excludes an ExtraTrees-specific explanation within the three audited families but does not establish external confirmation, universal model independence, deep-model portability, or best-backbone superiority.

Table 6: SHA-256 identifiers for the V375 portability audit.

Artifact	SHA-256
V375 pre-freeze JSON	ed6c7b6d119d4a3bb9111e27cae198c73f9c8590b3bb09756548ef06a95993e9
V375 audit program	d1924aa47c516e4f3474942e87044df7d5bec746531b8f3816244f610e92d5eb
V375 result report	862482438df32b59a19cfe8c4d15f413a11ee0baa2fd8e4451c9066e4d76562d
V375 independent verification receipt	26e81d0a73374a954e518dcda7e37bdd61ba04e88b407f8de55c82476a889376

6 Loss-geometry falsification record

The V366 extension was frozen before numerical execution. Its deterministic validator read no battery outcome and used seed 20,260,802 with tolerance 10^{-12} . It verified the metric identity, the exact bounded squared-loss supremum and feasible interval, active-boundary equality, adjacent exterior-point violation, recursive feasibility, and the full-real-line squared-loss divergence construction. All 27 aggregate checks passed. Across 200,000 random bounded configurations, the maximum formula discrepancy was 1.3323×10^{-15} . Across 2,000 non-increasing trajectories of length 128, the maximum budget excess was 7.7716×10^{-16} and the maximum output increase was zero. These tests target transcription and implementation errors; the analytical proofs remain the scientific basis of the claims.

Table 7: SHA-256 identifiers for the pre-frozen V366 loss-geometry extension.

Artifact	SHA-256
V366 loss-geometry pre-freeze	3e9b0494c23d10500889a27811fe8b9c29de74e60712271467e4d0c28ad0d1d0
V366 deterministic validator	6961dcf3d141397b07e6890d952d52e60689209a7386395d3f51e54e2557a344
V366 validation report	72cbb80861c9fc456a2aa7ba857757a33f3eb3585b7b9121366f70894476c8b0
V366 claim-source audit	4c430aca07747dc4651173c4458a69fc664d4fae4d684071fe0f97fe5dc2308a

7 Prefix nonexpansiveness falsification record

The V377 operator-stability protocol was frozen before numerical execution. The validator read no battery outcomes and used seed 20,260,802 with absolute tolerance 2×10^{-12} . It completed 3,888 exhaustive local protected-state checks, 21,168 exhaustive local projection checks, 108 proximal-equivalence checks, and 30,000 independently generated trajectory-pair checks. The maximum protected-state inequality excess was 1.1102×10^{-16} ; the maximum complete-output inequality excess was 3.2818×10^{-16} ; and the explicit witness attained the input separation 0.03 exactly up to floating-point representation. The report recorded a pass. These checks target implementation and proof-transcription failures; the theorem remains supported by the analytical proof in the main text.

Table 8: SHA-256 identifiers for the V377 operator-stability falsification.

Artifact	SHA-256
V377 deterministic validator	9abe330253305b02250bba00da337ec8003fe5c9aa99b535a4af2cb842e38b35
V377 validation report	2421dd474feb23a349b3be01354de240baf7b945ce5bbcdc95d5c7edbbd683f8
V377 four-test extension	66f2fc349826e6e9c1fa22529a99546e5a30405dbbc7ee11b4dc8f2c1c76f14e